

Names, data checks, and new variables

Placeholder names

Name	Stands for
dat, analysis	data frames
y	outcome
x, x1, x2, x3, z	numerical predictors
group	categorical predictor
m, m_multi, ...	fitted models
n, k, G	number of observations, slopes, groups

Read and inspect the data

```
dat <- read.csv("data.csv")

dim(dat)           # number of rows and columns
nrow(dat)          # number of rows
names(dat)         # variable names
head(dat)          # first six rows
summary(dat)       # summary of each variable
table(dat$group)   # count per category
mean(dat$y)        # mean
sd(dat$y)          # standard deviation
cor(dat$x, dat$y)  # correlation
tapply(dat$y, dat$group, mean) # mean of y per category
```

New variables and the analysis sample

```
dat$x_1k <- dat$x_euros / 1000 # in 1,000 euros
dat$group <- factor(dat$group) # categorical
dat$d <- as.integer(dat$group == "A") # 1 if A, else 0

# rows that meet the conditions
analysis <- subset(dat, x > 0 & group == "A")
```

Lecture 1: Simple linear regression

Population model and fitted line

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i.$$

OLS estimates in simple regression

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{s_{XY}}{s_X^2} = r_{XY} \frac{s_Y}{s_X}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Residual and sum of squared residuals

$$e_i = Y_i - \hat{Y}_i, \quad \text{SSE} = \sum_{i=1}^n e_i^2.$$

Change of the units of measurement by a factor c

Multiplied by c	New intercept	New slope
outcome only	$c \hat{\beta}_0$	$c \hat{\beta}_1$
predictor only	$\hat{\beta}_0$	$\hat{\beta}_1 / c$
outcome and predictor	$c \hat{\beta}_0$	$\hat{\beta}_1$

Fit a simple regression

```
m <- lm(y ~ x, data = dat)
coef(m)           # estimates
summary(m)        # coefficient table and fit
fitted(m)         # fitted values
residuals(m)      # residuals
```

Scatter plot with the fitted line

```
plot(y ~ x, data = dat,
     xlab = "Predictor (unit)", ylab = "Outcome (unit)")
abline(m, col = "blue", lwd = 2)
```

Fitted values for chosen values x_A and x_B of the predictor

```
b <- coef(m)
b[1] + b[2] * x_A # fitted value by hand

new_cases <- data.frame(x = c(x_A, x_B))
predict(m, newdata = new_cases) # fitted values
```

Lecture 2: What the model assumes and how residuals reveal it

Residual plot and normal Q-Q plot

```
plot(fitted(m), residuals(m),
     xlab = "Fitted value", ylab = "Residual")
abline(h = 0, lty = 2)
```

```
qqnorm(rstandard(m))
qqline(rstandard(m))
```

Standardised residuals and Cook's distance

```
summary(residuals(m)) # summary of the residuals
rstandard(m)          # standardised residuals
cooks.distance(m)     # Cook's distances
nobs(m)               # number of observations n
```

Flagged observations

$$|\text{standardised residual}| > 3, \quad \text{Cook's distance} > \frac{4}{n}.$$

```
sum(abs(rstandard(m)) > 3) # how many beyond ±3
sum(cooks.distance(m) > 4 / nobs(m)) # how many above 4/n
```

Lecture 3: Coefficient inference, fit, and the F-test

Two-sided test of one coefficient

$$H_0 : \beta_j = 0, \quad H_1 : \beta_j \neq 0, \quad t = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}.$$

Reject H_0 at the 5% level if $|t_{\text{obs}}| \geq t_{0.975, df}$.

Degrees of freedom and 95% confidence interval, model with k slopes and an intercept

$$df = n - k - 1, \quad \hat{\beta}_j \pm t_{0.975, df} \text{SE}(\hat{\beta}_j),$$

simple regression: $k = 1$, $df = n - 2$.

```
summary(m)           # coefficient table and fit
confint(m, level = 0.95) # 95% confidence intervals
df.residual(m)        # residual degrees of freedom
qt(0.975, df.residual(m)) # critical value, 95% interval
```

Labels in the output of summary(m)

Label in the output	Symbol
Estimate	$\hat{\beta}_j$
Std. Error	$\text{SE}(\hat{\beta}_j)$
t value	t
Pr(> t)	two-sided p-value for $H_0 : \beta_j = 0$
Residual standard error	s , with $df = n - k - 1$
Multiple R-squared	R^2
Adjusted R-squared	R_{adjusted}^2
F-statistic	overall F , with k and $n - k - 1$ degrees of freedom

The t statistic and the two-sided p-value step by step

```
tab <- summary(m)$coefficients
t_value <- tab["x", "Estimate"] / tab["x", "Std. Error"]
p_value <- 2 * pt(-abs(t_value), df = df.residual(m))
```

Sums of squares and R^2

$$\text{SST} = \sum_i (y_i - \bar{y})^2, \quad \text{SSE} = \sum_i (y_i - \hat{y}_i)^2,$$

$$\text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2,$$

$$\text{SST} = \text{SSR} + \text{SSE}, \quad R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{\text{SSR}}{\text{SST}}.$$

```
y <- dat$y # outcome values
yhat <- fitted(m) # fitted values
```

```
SST <- sum((y - mean(y))^2)
SSE <- sum(residuals(m)^2)
SSR <- SST - SSE
R2 <- 1 - SSE / SST
```

Residual standard error and adjusted R^2

$$s = \sqrt{\frac{\text{SSE}}{n - k - 1}}, \quad R^2_{\text{adjusted}} = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}.$$

```
summary(m)$r.squared      # R squared
summary(m)$fstatistic     # overall F, k, and n - k - 1
```

Overall F-test, model with k slopes

$$H_0 : \beta_1 = \dots = \beta_k = 0,$$

$$H_1 : \text{at least one of } \beta_1, \dots, \beta_k \text{ is not } 0,$$

$$F = \frac{\text{SSR}/k}{\text{SSE}/(n - k - 1)}, \quad \text{simple regression: } F = t^2.$$

Heteroskedasticity-robust standard errors (HC1)

```
library(lmtest)
library(sandwich)
```

```
# robust standard errors, t statistics, and p-values
coeftest(m, vcov. = vcovHC(m, type = "HC1"))
# robust 95% confidence intervals
coefci(m, vcov. = vcovHC(m, type = "HC1"))
```

Lecture 4: Multiple regression and what controls do**Model with k predictors**

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \varepsilon_i.$$

```
m_multi <- lm(y ~ x1 + x2 + x3, data = analysis)
summary(m_multi)      # coefficient table and fit
confint(m_multi)      # 95% confidence intervals
```

Fitted difference for changes $\Delta x_1, \dots, \Delta x_k$ of the predictors

$$\Delta \hat{Y} = \hat{\beta}_1 \Delta x_1 + \dots + \hat{\beta}_k \Delta x_k.$$

Fitted difference between two cases A and B

```
cases <- data.frame(
  x1 = c(x1_A, x1_B),
  x2 = c(x2_A, x2_B),
  x3 = c(x3_A, x3_B)
)
fits <- predict(m_multi, newdata = cases) # fitted values
fits[2] - fits[1]                        # B minus A
```

Regression table with one column per model

```
# install once: install.packages("modelsummary")
library(modelsummary)
m_a <- lm(y ~ x1, data = analysis)
m_b <- lm(y ~ x1 + x2, data = analysis)
modelsummary(
  list("(1)" = m_a, "(2)" = m_b, "(3)" = m_multi),
  stars = c("*" = 0.05, "***" = 0.01, "****" = 0.001),
  gof_map = c("nobs", "r.squared", "adj.r.squared")
)
```

Lecture 5: Categories as predictors and joint tests**Reference category and model with a categorical predictor**

```
analysis$group <- factor(analysis$group)
# categories: the first one is the reference
levels(analysis$group)
analysis$group <- relevel(analysis$group,
  ref = "reference")
```

```
m_group <- lm(y ~ x1 + x2 + group, data = analysis)
summary(m_group)
coef(m_group)
confint(m_group)
```

The same model with another reference category

```
analysis$group_alt <- relevel(analysis$group,
  ref = "other")
m_alt <- lm(y ~ x1 + x2 + group_alt, data = analysis)
# TRUE if the fitted values of the two models are equal
all.equal(fitted(m_group), fitted(m_alt))
```

Partial F-test of nested models

```
reduced <- lm(y ~ x1 + x2, data = analysis)
complete <- lm(y ~ x1 + x2 + group, data = analysis)
anova(reduced, complete)
```

Complete (unrestricted) model: k_C slopes. Reduced (restricted) model: the complete model without q of its terms, with the same outcome and the same observations.

H_0 : the q coefficients left out of the reduced model are all 0,

$$F = \frac{(\text{SSE}_R - \text{SSE}_C)/q}{\text{SSE}_C/(n - k_C - 1)}.$$

Labels in the output of anova(reduced, complete)

Label in the output	Symbol
Res.Df	$n - k - 1$ of each model
RSS	SSE_R in row 1, SSE_C in row 2
Df	q
Sum of Sq	$\text{SSE}_R - \text{SSE}_C$
F	partial F
Pr(>F)	p-value of the partial F-test

One-way ANOVA: one model, G groups

```
m_anova <- lm(y ~ group, data = analysis)
anova(m_anova) # ANOVA table
qf(0.95, df1, df2) # critical value of F for a 5% test
# one-way ANOVA: df1 = G - 1, df2 = n - G
# partial F-test: df1 = q, df2 = n - k_C - 1
```

$$H_0 : \text{all } G \text{ group means are equal}, \quad F = \frac{\text{SSR}/(G - 1)}{\text{SSE}/(n - G)}.$$

Labels in the output of anova(m_anova)

Label in the output	Symbol
Df	$G - 1$ (group), $n - G$ (Residuals)
Sum Sq	SSR (group), SSE (Residuals)
Mean Sq	Sum Sq / Df
F value	F
Pr(>F)	p-value of the F-test

Lecture 6: Interactions and curvature**Interaction of a numerical predictor X with a dummy D**

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 D + \hat{\beta}_3 XD.$$

Group	Intercept	Slope
$D = 0$	$\hat{\beta}_0$	$\hat{\beta}_1$
$D = 1$	$\hat{\beta}_0 + \hat{\beta}_2$	$\hat{\beta}_1 + \hat{\beta}_3$

```
m_interaction <- lm(y ~ x * group + z, data = analysis)
# x * group expands to x + group + x:group
summary(m_interaction)
```

Interaction with a numerical moderator Z

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 Z + \hat{\beta}_3 XZ.$$

$$\text{Slope of } X \text{ at } Z = z : \quad \hat{\beta}_1 + \hat{\beta}_3 z.$$

```
m_moderator <- lm(y ~ x * z, data = analysis)
# x * z expands to x + z + x:z
summary(m_moderator)
```

Centred predictor

```
# x minus its mean
analysis$x_c <- analysis$x - mean(analysis$x)
m_centred <- lm(y ~ x_c * group + z, data = analysis)
```

Quadratic model

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2.$$

$$\text{Fitted change from } x \text{ to } x + 1 : \quad \hat{\beta}_1 + \hat{\beta}_2(2x + 1).$$

```
m_linear <- lm(y ~ x + z, data = analysis)
m_quadratic <- lm(y ~ x + I(x^2) + z, data = analysis)
# I(x^2) is the square of x
```

```
summary(m_quadratic)
anova(m_linear, m_quadratic)
```

Commands by purpose

Purpose	R command
Fit a model	<code>lm(y ~ predictors, data = dat)</code>
Coefficient table and fit	<code>summary(m)</code>
Coefficients only	<code>coef(m)</code>
95% coefficient intervals	<code>confint(m)</code>
Fitted values and residuals	<code>fitted(m), residuals(m)</code>
Summary of the residuals	<code>summary(residuals(m))</code>
Standardised residuals	<code>rstandard(m)</code>
Cook's distance	<code>cooks.distance(m)</code>
Fitted values for new cases	<code>predict(m, newdata = ...)</code>
R^2 , adjusted R^2 , overall F	<code>summary(m)</code>
Residual degrees of freedom	<code>df.residual(m)</code>
Critical value of a 95% interval	<code>qt(0.975, df.residual(m))</code>
Partial F-test of nested models	<code>anova(reduced, complete)</code>
ANOVA table of one model	<code>anova(m)</code>
Reference category of a factor	<code>relevel(factor(group), ref = "category")</code>
Interaction	<code>x * group</code>
Square in a formula	<code>I(x^2)</code>
Robust standard errors	<code>coeftest(m, vcov. = vcovHC(m, type = "HC1"))</code>
Regression table with several models	<code>modelsummary(list(...))</code>